Label-Based Sorting Automation in the Logistics Industry Using Factory I/O

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ABSTRACT: The logistics sector is experiencing growing demands to improve both operational efficiency and precision in its production and distribution procedures. A potential remedy for optimizing these activities is the adoption of RFID (Radio Frequency Identification) technology aimed at automating sorting processes. This study outlines a detailed framework for a label-based sorting automation system that employs an RFID reader, incorporated within a simulated context utilizing Factory I/O, a prominent platform for industrial automation simulation. The proposed framework will use RFID tags to provide real-time tracking and object classification so that accurate identification and handling errors can be avoided. Our simulation demonstrates that this automated solution is feasible for reducing labour costs, increasing throughput, and better management of inventory. The results are an indication that RFID-based sorting automation has great potential to revolutionize logistics operations by making the supply chain architecture more efficient, responsive, and agile. In the dynamic world of the logistics industry, more efficient and accurate sorting is a fundamental requirement. RFID technology provides an innovative solution to the problem that is inherent in conventional sorting techniques. The problems associated with the manual process of sorting include delays, human error, and inefficiency in operations. With this technology, there is a way of ensuring that goods can have unique identification tags attached to them, allowing for real-time tracking and automatic sorting according to factors such as destination, type, or priority.

This study presents the future implications of RFID-enabled sorting automation for changing logistics operations. Factory I/O simulation is of significant value as a source of information for stakeholders on how RFID technology influences activities in sorting, thus drawing lessons applicable to real scenarios. With the adoption of digital transformation by the logistics sector, these results of this research also contribute to the larger discourse on how supply chain management can be improved through such innovative technological approaches.

Keywords: - Factory I/O, RFID Reader, RFID Reader-based Sorting

1. INTRODUCTION

Since economics has always played a significant role in the development of industry, Low-Cost Automation (LCA) has to be developed in order to accurately sort these products. The secret to success in the automation sector is constant innovation and the discovery of practical methods to increase output and reduce operating costs. Increased market knowledge and a strategic reassessment of the value chain are required due to the growing demand for automation systems. The primary goals of industrial automation are to create automations that are inexpensive, long-lasting, require minimal maintenance, and are as user-friendly as feasible.

Factory I/O is a highly interactive simulation tool for teaching automation and control systems. It enables students, engineers, and hobbyists to create and mimic industrial processes in a secure virtual environment. The logistics sector forms the backbone of the world's economy since it exercises control over the distribution of products directly from producers to end-users. While consumer demands are changing into expedited deliveries and more accurate inventory management, logistics companies are now on an increasing quest for automated solutions to conquer these challenges. Traditional sorting methods based on manual labor, scanning, and barcodes often lead to inefficiencies, errors, and escalated operational costs. Given the above problems, Radio Frequency Identification technology offers a very attractive solution. RFID allows objects to be identified and followed wirelessly through the exchange of data between tags and readers via electromagnetic fields. In terms of advantages, the technology offers real-time visibility, faster scanning speeds, simultaneous reading of several items without bottlenecks in the sorting process, and is highly efficient and automated as it works well with a factory automation system that calls for minimal human intervention during the sorting process.

The current paper discusses the potential use of RFID reader-based sorting automation in the logistics sector and evaluates the practicality of introducing such automation. Factory I/O is a simulation tool built to mimic the procedure and process of industrial automation. The following investigation will sketch the practical considerations relevant to the implementation of RFID technology in logistics, including its system architecture and issues surrounding the implementation process. Based on this, the motivation for this study is to provide a framework for logistics firms considering the implementation of RFID-based automation solutions that will, in turn, help upgrade operational efficiency and maintain their competitive advantage in an ever-changing environment. Sorting by label needs the integration of control systems that guide things depending on their labels. The simulation may include developing serial number networks that interact with the sorting system to accurately guide the objects. Sorting systems in the logistics sector are in charge of effectively guiding items to designated locations, frequently using information stored in printed labels, RFID tags, or barcodes. Sorting and handling of merchandise have to be efficient enough for the operational efficiency of businesses. With increasing volumes to be processed, conventional methods of sorting are lagging behind in meeting the emerging needs for speed and precision in handling goods. Transformative possibilities in optimizing the workflows of logistics are seen through label-based automation offered by RFID technologies.

RFID readers allow real-time tracking and product identification through scanning of affixed RFID tags on commodities. Information that is pertinent to the identification of a product and locational identifiers in the tag allows the sorting of items into other categories or locations, and will ensure that more efficiency in this process can be attained by integrating RFID technology and control systems within logistics operations for better sorting and routing products to desired destination locations.

2. LITERATURE REVIEW

'A Comparative Efficiency Simulation of Package Sorting Systems using Factory I/O' This is the student project that shows two of a possible package-sorting configuration modelled and compared in the Factory I/O 3D simulation environment. The authors use control logic PLC (Control I/O fb) and connect a virtual HMI / SCADA in Allen-Bradley CompactLogix (RS Logix 5000 / FactoryTalk). The reason is that the study emphasizes that virtual commissioning in the Factory I/O allows the ability to compare throughput and control robustness of topologies before dedicated hardware commits [1]. The present paper puts forward the idea of Low-Cost Automation (LCA) solutions to sort out the streams of products predominantly similar (small differences in height, weight, color, or shape). It concentrates on cost-effective sensor-actuator selection and simplified PLC logic that will achieve acceptable line speed accuracy - an option that seeks to solve the challenge of capital cost and maintainability in small/medium enterprises environments. The work focuses on trade-offs between maintainability, cost, and accuracy [2]. Such a survey places the topic of conveyor-based sortation within the context of an operational-research (OR) problem: routing, buffering, sequencing, capacity planning, and throughput optimization. It integrates mathematical models, algorithms, and criteria of a design and assessment of high-speed sortation conveyors, offering advice on trade-offs (e.g., buffer size versus throughput) and proportions that ought to guide mechanical control logic and physical configuration. The implementational focus of the studies is complemented by the OR perspective that provides formal optimum planning tools to the system planners [3].

Alaameri et al. provide a workable example of the color sorting system that includes PLC programming (TIA Portal), Factory I/O simulation, and industrial color-sensing devices. Sensor location and chute activation logic, along with simulation verification, are described in the paper to demonstrate that the design works reliably to differentiate color and synchronize timing when multiple chutes are used to make sorts. It shows that Factory I/O can be employed together with TIA Portal to test ladder/FB-code before making a deployment. [4]. This paper refers to height measurement as a light and powerful capability of sorting; lasers or ultrasonic sensors are used to measure the height of the objects in a conveyor belt, and PLC logic tags objects under height bands to route the item. Simulation of sensor read timing, conveyor spacing, and actuation timing is performed, using integration with Factory I/O to make sure separation at speed is mathematically correct. Sensor calibration, debounce/averaging techniques, and PLC timing of high-throughput lines are emphasized in the study.[5]. The work uses a Factory I/O model of an airport baggage handling layout and implements ladder logic with WinSPS-S7 (Siemens S7 family). It touches on flight-based routing, delayed baggage handling, and multi-flow merging/splitting, and how timing conflict and belt-synchronization problems can be identified using virtual testing. The paper highlights the fact that Factory I.O. is well suited to complex and multi-path systems, as well as those appropriate in pre-commission training of control engineers intentionally prior to live commissioning. [6]. A facility and engineering reference including a crash course on conveyors, sorting equipment, buffering,

storage, and ergonomics of material handling systems. It provides design equations, lists equipment selection, and provides design instruction on how to design flexible, high-throughput material handling systems- a crucial background reference that informs layout choices and mechanical trade-offs in the design of an automated sorting system [7]. Rieger and Kalus survey computer-vision approaches to industrial automation with emphasis upon object-detection, classification, and localization problems. They describe classical image processing and recent neural-network solutions with a focus on issues of robustness (lighting invariance, occlusion, domain shifts) and two-stage detectors (classification, localization). The article presents the role of vision to enhance the control of PLC in complex sorting applications, but also presents problems of data-labeling, computing, and deployment in real-time systems. [8].

In this handbook, authoritative chapters are gathered on robotic manipulation, motion planning, sensing, and system integration. It is related to sorting in robot-based pick-and-place, perception-based manipulation, and systems design at the logistics cell level. The literature gives the theoretical and practical guidelines for robot applications in sorting lines, mainly in handling operations where flexibility, non-contact handling of products, or object geometry is needed [9]. Rybczak et al. propose a 3D simulation method that connects the Factory I/O with TIA Portal and the TIA S7-1214 controller (controller). These give visualization, Ladder logic examples, as well as the real-time exchanges between the virtual plant and PLC code. Their findings reaffirm the pedagogical and prototyping cost of a dynamic programming environment and show better debugging and validation processes when control code is validated in a real simulated environment [10]. The textbook by Groover gives the principles of production systems, material handling, automation technologies, and integration of the systems. In the case of sorting systems, it develops conceptual frameworks about process flow, inventory/ queueing effects, and how to use identification technologies (barcode/RFID) in routing decision making, which is a significant theoretical background upon which it is important to place PLC-level control into its group perspective of manufacturing and logistics.[11]. This work discusses the practical problem of missing annotated industrial datasets in supervised ML to be used in automation. It revises approaches (crowdsourcing, active learning, semi-supervised learning) and concludes that industrial examples continue to rely on manual annotation, due to domain complexity and due to the low-error requirements for labeling, rendering automated interventions unreliable. The authors of the paper urge the availability of unified labeling pipelines and simulation-to-real transfer solutions to decrease the cost of annotations of vision-based sorting [12].

The Logistics Management. In brief, this source presents an overview of identification technologies in logistics (barcode scanning and RFID) and how these inform routing choices in an automated environment. It presents a survey of reader/tag technologies, information coding methods, and how connected to PLC/MES layers for deterministic sorting and tracking, and notes that in terms of suitability to the parcel and pallet-level sorting within logistic centers, the unique identity technology still plays a pivotal role [13]. Prodan and Bostan address several industrial PLC applications, pointing out the strengths of PLC such as deterministic timing, troubleshooting capability, modular I/O, and a long field-reliability record. They provide case studies on the use of PLC-based control in sorting and conveyance processes and propose that PLCs will continue to form the foundation of industrial control, whilst suggesting future research studies on the limitations of PLCs as they continue to evolve into more complex, AI-aided types of systems. [14]. PLC PLC-based color sorting and pick-n-place system that was supposed to lessen the manual color sorting of packaging lines. It focuses on ease of use, ease of use and cost-effectiveness, demonstrating sensor choice, actuator timing, and elementary ladder logic, an automated color sorting and pick-and-place cycles to match the needs of an SME [15]. Li and Song conduct research into automated label recognition (vision or optical character recognition) and pneumatic actuation on high-throughput sorting. They present the use of pneumatic pushers and controlled valve timing as effective in fast mechanical sorting and the potential to fulfill packaging industry requirements since trusted labels and deterministic actuation can be used. As said in the paper, there exists a need to have strong pre-processing and mechanical timing to make low-error and high-speed sorting in production setups possible [16].

3. METHODOLOGY

The challenge of always achieving logistics efficiency is primarily centered on good sorting, something that has always traditionally been carried out by very slow manual methods requiring a large amount of labor. The traditional methods not only take much time but also pose a tremendous risk for errors to occur while sorting goods. This means that the process results in higher operational costs and consequently leads to delivery times being delayed, thereby affecting customer satisfaction. With this in mind, this project aims to automate the process of sorting by using RFID technology within a simulated logistics environment created using Factory I/O software. The major objective of this project is to design a highly advanced automated sorting system that can correctly sort five different types of products based on their unique RFID tags. This would itself be an innovative system with a conveyor design featuring five specialized pushers and sophisticated sensors so that the products can be sorted with high precision and efficiency. The design should reduce as much manual labor as possible while trying to

increase overall throughput and efficiency significantly.

In the present work, an RFID-based automated sorting system has been designed for five different categories of products in the context simulated by Factory I/O. The setup used a rectangular conveyor belt to simulate the movement of goods that, in the future, may allow the appropriate placement of sorting equipment. All of five products were fitted with unique RFID tags. Along the conveyor line, it is indeed true that an RFID reader was placed so strategically to scan those tags whenever products pass through its range. The output of this RFID reader is fed into a control system that processes the information. This control system could be a simulated PLC or any variety of logic controller available in the Factory I/O environment. Furthermore, five virtual pusher mechanisms have been wisely provided along the length of the conveyor, and these have been placed carefully at such predefined intervals to ensure best functionality. Each of the individual mechanisms was related to a particular type of product and was triggered by signals emanating from the control system. In addition, a virtual diffuser sensor was mounted on the side of each pusher to sense the presence of a product at the assigned sorting position. This configuration suggests that the mechanisms of the pushers activate only when a product is sensed to exist and in no other situation. This is the control program, by the way, using ladder logic or an appropriate form of programming, efficiently managing and orchestrating the whole sorting process. It receives and processes signals coming from the RFID reader to align them properly with the relevant pusher and, while at it, takes charge of managing the signals from the diffuser sensors. Ultimately, in the conclusion of this integration, virtual sensors, actuators, and the RFID reader within the Factory I/O application were successfully incorporated and synchronized with the control program to enhance overall functionality and efficiency. In the operational process, virtual items embedded with different RFID tags, which are unique identifiers used for tracking, were placed strategically on the conveyor system for effective movement and handling. As these items moved smoothly past the RFID reader, their tags were efficiently scanned, capturing important information, and the resulting data was subsequently sent to the control system to thoroughly evaluate the appropriate sorting action that was required for each item based on the scanned information. The moment the good entered the sorting zone, the diffuser sensor assigned to that particular good sent a signal to the control system, which indicated that this particular type of good had just entered the sorting zone; in response to such a signal, the control system would then activate the proper pusher that was made to deflect the product off of the moving conveyor belt. The entire sorting system operated continuously and thus efficiently sorted the various types of virtual goods that were present throughout the entire simulation process.

4. RESULT AND DISCUSSION

The RFID-based automatic sorting system, developed and implemented within the Factory I/O simulation environment, brought about a highly remarkable improvement in the efficiency of the logistics operations as compared with the manual sorting procedures that were being conducted traditionally. From the point of view of performance during operation, this new automatic system proved to be an efficient and reliable method of classifying five different products distinguished by their respective RFID tags that were attached to the products. This not only eliminated any probable error that could occur in the operational process but also contributed to a considerable saving of time for the task of sorting.

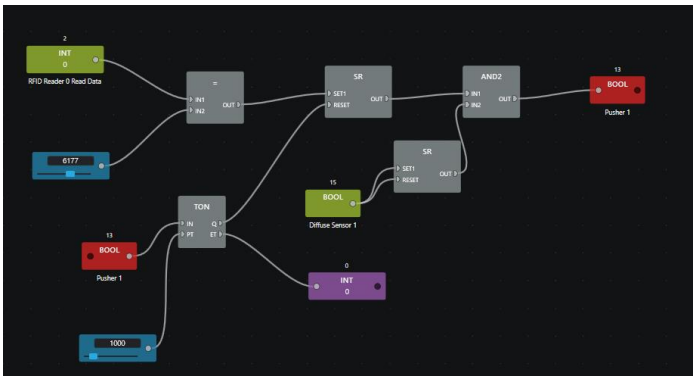

Fig.1. Control I/O program for Label-based Sorting

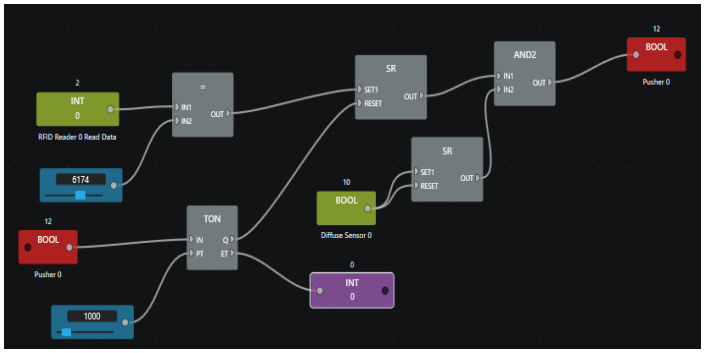

Fig.2. Control I/O program for Label-based Sorting

This enabled the well-designed rectangular conveyor layout to allow an unbroken and seamless flow of virtual goods along its length, thus ensuring that the process was smooth and efficient. Secondly, the placement of the RFID reader was not just accidental; it was instead strategic because it ensured that tags were constantly being scanned in real-time as products moved down the conveyor. In this way, when a particular product approached the RFID reader, the critical information transmitted regarding that particular product could be immediately transmitted to the control system for further processing and analysis. This high-level skill and expertise that the control system employed by using ladder logic programming techniques successfully managed the numerous signals by judiciously connecting them with the unique pusher mechanisms that were meant to satisfy the distinct needs of each good type. Such a careful integration of parts not only allows for speed in making decisions about sorting but also ensures that these decisions are made with unprecedented accuracy, which would enhance the efficiency of sorting.

Fig.3. Simulation of Label-Based Sorting

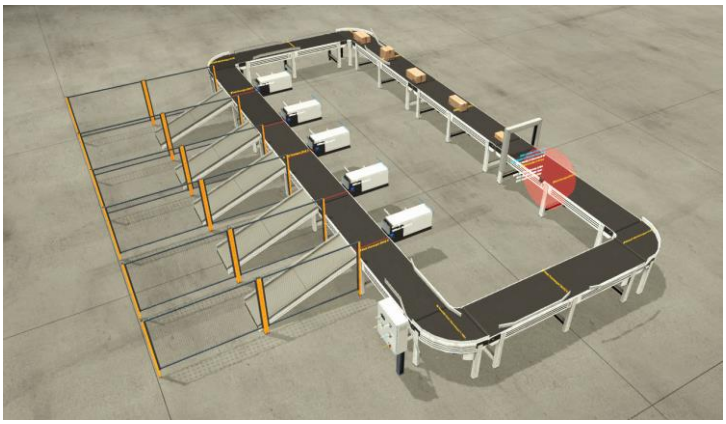
Fig.4. Simulation of Label-Based Sorting

The mechanisms that constitute the virtual pusher, along with diffuser sensors, have successfully and significantly reduced the spurious activations to occur. This excellent performance ensures that the virtual pusher is involved only when a product is being detected within the specific sorting area. Because of this particular feature, there has been a significant reduction in the chances of mis-sorting, which is one of the most important and most common issues faced by manual sorting systems. The system's ability to work continuously also allows it to process an exceptionally high volume of items in an orderly and systematic manner. Therefore, there has been an enormous improvement in both throughput and delivery speed, thus enhancing overall performance. In

addition, the removal of manual labor further contributed to lowering the operational costs and, at the same time, reducing the possibility of human error. Hence, the entire process concerned with item sorting became significantly more reliable and efficient. Likely, customers will also experience greater satisfaction as a result of increased processing times and decreased delivery times, benefits that this automated system is likely to offer.

The system was rigorously tested and assessed with various performance metrics to understand it better. Some of the most critical metrics used during assessment included sorting speed, accuracy, and operation efficiency. All three are extremely crucial in judging the outcome delivered by any sorting system. The test results indicated that an automatic system could indeed sort vastly larger numbers of items per hour than similar conventional methods, but without much compromise on the sort accuracy level. The integration of Radio Frequency Identification, which is often referred to as RFID technology, has offered the facility of implementing real-time tracking along with more advanced inventory management capabilities that can be utilized efficiently; this development is capable of improving true logistics applications by managing the stock levels and resource usage efficiently.

5. CONCLUSION

The devised RFID-based automated sorting system, integrated with the Factory I/O simulation framework, has been proven to produce significant improvements in logistics efficiency when compared to traditional manual techniques for sorting, which have been in use for decades. The system effectively categorizes five different types of products by minimizing the existing inaccuracies associated with sorting and the effective time spent on these processes. The strategic placement of the RFID reader also facilitated instantaneous transfer of data, giving way to timely and accurate decision-making during the sorting operations, thereby making the whole process more efficient.

This automation reduces dependency on human labor, hence reducing the operational costs and the chances of human error that may be contributed by human interference during several processes. Additionally, this system allows continuous running with no breaks or downtime, which increases the overall throughput and delivery rates, thus boosting efficiency in all operations. After carrying out an elaborate analysis of the performance measures, which are the speed of sorting, accuracy, and operational efficiency, it was evident that the automated sorting system is superior while maintaining a high degree of accuracy in its operations compared to the traditional method.

Further, the smooth interface of very advanced control programming with state-of-the-art sensor technology is critical to an incredibly efficient and remarkably sophisticated sorting mechanism that is accurate and dependable. This significant increase in logistical capacity will indeed lead to optimized operational workflows, and such optimized workflows are indeed expected to boost customer satisfaction significantly through substantially faster processing and delivery times. In a nutshell, the outcomes of this study point towards the fact that revolutionary capability is emerging in the contemporary management of supply chains. This capability is enabled through the effective utilization of RFID technology. The new advancement, therefore, hints at substantial efficiency improvements and increased accuracy and responsiveness in performing a range of logistics activities.

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